

SDP 302

Assignment: OLS & Multicollinearity

Professor: Michael Wegener | Term: Fall 2026 |

Instructions & Information

The Iris flower dataset is a classic multivariate dataset introduced by Fisher [1936](#). Collected by botanist Edgar Anderson from the Gaspé Peninsula, it contains measurements for 150 Iris flowers spanning three distinct species. Although relatively small by modern standards, it serves as a quintessential benchmark in data science for pattern recognition, classification, and data visualisation. Consequently, the Iris dataset is built into nearly all major data analytics platforms. Standard textbooks, including James et al. [2013](#), frequently use it to demonstrate foundational techniques in data science and machine learning.

References

- Fisher, R. A. (1936). “The Use of Multiple Measurements in Taxonomic Problems”. In: *Annals of Eugenics* 7, pp. 179–188.
- James, Gareth et al. (2013). *An Introduction to Statistical Learning*. Vol. 103. Springer Texts in Statistics. New York, NY: Springer New York.

[EXR] Exercise 1 (OLS & Multicollinearity: The Iris Data Set)

The Iris dataset is available directly within Python’s `seaborn` library and can be loaded using `seaborn.load_dataset("iris")`. Import the dataset into a `pandas.DataFrame` and answer the following questions:

- For each species fit a simple linear regression model via OLS to predict `sepal_length` using `petal_length` as the sole predictor variable.
- Report the coefficient of determination R^2 . What proportion of the total variance in sepal length is explained by petal length?
- Conduct a t -test for the null hypothesis $H_0 : \beta_1 = 0$ at the significance level $\alpha = 0.05$. State the test statistic, the resulting p -value, and your conclusion.
- Extend the framework to a multiple linear regression model predicting `sepal_length` using `petal_length`, `sepal_width`, and `petal_width`. Estimate the parameters via OLS and present the summary table.
- Evaluate the model for potential multicollinearity by calculating the variance inflation factor for each explanatory variable. Identify any variables exhibiting severe collinearity.